http://www.csvtuonline.com

Printed Pages – 5+3=8 Roll No. :

224314(28)

024514(28)

Diploma in Engg. (Third Semester) Examination, Nov.-Dec., 2016

(Old Scheme)

(Elect. Engg. Branch)

BASIC ELECTRONICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *All questions are compulsory. Draw the diagram neatly. Internal choices are given in questions.*

1. Choose the correct answer : 5×2=1

(i) To operate a transistor in ACTIVE region gase-emitter junction and collector base junction should be respectively.

http://www.csvtuonline.com

http://www.csvtuonline.com

(a) Farward bias, Reverse bias

(b) Farward bias, Farward bias

(c) Reverse bias, Farward bias

(d) Reverse bias, Reverse bias

(ii) Efficiency of fullwave rectifier is :

(a) 40%

(b) 81·2%

(c) 50%

(d) 100%

(iii) In Voltage shunt feedback amplifier the input resistance and output resistance will be respectively :

(a) Increase, Decrease

(b) Decrease, Decrease

(c) Decrease, Increase

(d) Increase, Increase

(iv) In parallel LC resonance, circuit impedence is :

(a) L/CR

(b) R/LC

(c) CR/L

(d) C/LR

http://www.csvtuonline.com

http://www.csvtuonline.com

(v) PIV in a center-tapped rectifier is :

(a) 2 Vm

(b) Vm

(c) 0.5 Vm

(d) Zero

2. (a) Explain the concept of barrier potential in P-N junction. 5

(b) Why common emitter configuration is most popular configuration in transistor. Also prove that in a transistor : 5

$$\beta = \frac{\alpha}{1-\alpha}$$

(c) Compare zener diode with simple diode. 5

3. (a) Explain zener diode as a shunt regulator with diagram. 5

Or

Describe negative clipper circuit with diagram.

(b) Explain bridge rectifier on following points : 10

(i) Circuit diagram

http://www.csvtuonline.com

http://www.csvtuonline.com

(ii) Efficiency

(iii) Ripple factor

4. (a) Draw circuit of emitter follower and explain its working. 5

(b) Draw the circuit diagram of single stage transistor amplifier and explain the function of each component. 10

Or

Draw and explain block diagram of four types of feedback. Give the advantages of negative feedback in amplifier.

5. (a) Explain working of transformer coupled amplifier with circuit diagram and frequency response curve. 10

Or

Explain working of RC coupled amplifier with circuit diagram and frequency response curve.

(b) Explain Barkhausen criteria. 4

6. Explain the working of single tuned voltage amplifier with circuit diagram. Also describe its frequency response and limitations. 10

http://www.csvtuonline.com

http://www.csvtuonline.com

7. Describe Hartley oscillator with circuit diagram. 6

Or

Describe Weinbridge oscillator with its circuit diagram.

8. Write short notes on : 15

(any **three**)

(i) Bistable multivibrator

(ii) Clamper

(iii) π -filter

(iv) Staggered tuning

(v) Complete DC power supply

http://www.csvtuonline.com

http://www.csvtuonline.com

http://www.csvtuonline.com

http://www.csvtuonline.com

http://www.csvtuonline.com